ADRI RAJARAMAN

(508)410-8510 arajaraman@wpi.edu Research Portfolio Cambridge, MA

Education

Worcester Polytechnic Institute, MA
Master of Science in Robotics Engineering *Aug 2021 – Dec 2023*

Manipal University, India
Bachelor of Technology in Electronics Engineering (Minor in Embedded Systems) *Aug 2015 – Jun 2019*

Publications

- K. Bonofiglio, W. Wang, E. R. Wilke, **A. Rajaraman**, and L. Fichera, "A Starter's Kit for Concentric Tube Robots," arXiv preprint arXiv:2407.06943, 2024.
- **A. Rajaraman** et al., "Acoustic 3D localization of a piezoelectric receiver using the imaging beams of an unmodified ultrasound probe," manuscript in preparation.

Research Experience

Brigham & Women's Hospital, Harvard Medical School | Research Volunteer *Sep 2025 – Present*

- **Ultrasound-Guided Needle Tracking** (under *Prof. Jayender Jagadeesan*)
- Architected an end-to-end acoustic 3D localization system that estimates the position of a piezoelectric receiver relative to an unmodified 210-element curvilinear ultrasound array from the time of flight of the probe's own imaging beams, validated against electromagnetic tracking ground truth.
- Designed and integrated a three-stream synchronized acquisition system (oscilloscope via SCPI, EM tracker, IMU over a custom serial protocol) with fail-soft error handling for unattended 300-trial capture sessions; derived data-quality verification gates from acoustic physics.

Harvard Medical School | Research Intern *May 2023 – Aug 2023*

- **Magnetically Actuated Suturing Device** (under *Prof. Nobuhiko Hata*)
- Designed and built a prototype electromechanical device to actuate magnetic suturing needles; fabricated custom electromagnets, modeled field intensity in MATLAB, and developed closed-loop feedback-linearization control validated via real-time needle tracking against camera and fluoroscopic imaging on tissue phantoms.

Worcester Polytechnic Institute | Research Assistant *Sep 2021 – Dec 2023*

- **Concentric Tube Robots for Minimally Invasive Surgery** (under *Prof. Loris Fichera*)
- Built an open-source *robot and framework* using additive manufacturing for patient-specific devices; implemented inverse kinematics via Euler-Bernoulli beam equations and energy minimisation, validated against stereo-camera motion capture.
- Co-authored an open-source educational kit for concentric tube robots (about $700 in materials) with self-study materials, evaluated in a four-week human-subjects study with first-year graduate students (arXiv:2407.06943).
- **SpaceGoat** (NASA BIG Idea Challenge, 2022): a quadruped with a novel transmission using a centralised actuator that decouples input torque and speed per degree of freedom.

Indian Institute of Science (IISc) | Bachelor's Thesis and Research Assistant *Jan 2019 – Feb 2021*

- **Ultrasound Probe Pose Estimation for 3D Reconstruction** (bachelor's thesis, under *Prof. Manish Arora*; multi-institutional project with Cranfield University, UK, on remote obstetric ultrasound)
- Designed a mechanical arm to track the pose of a handheld ultrasound probe, and explored a deep-learning framework that estimates probe pose from changes between consecutive ultrasound images, toward 3D reconstruction of the scanned region.
- **Wearable Motion Capture and Musculoskeletal Assessment System:** designed and deployed a wearable full-body inertial motion-tracking system with real-time Extended Kalman Filter state estimation and gait calibration; wrote embedded software in Embedded C and C# under real-time and wireless constraints.

Industry Experience

SharkNinja | Senior Electrical Systems Engineer (2026–), Robotics Engineer (2024–26), Robotics Intern (2022)

- **Depth Estimation with a Single Camera:** zero-shot transfer for relative depth by creating a dense point cloud for object detection; developed a sensor-fusion algorithm using an RGB camera and 2D LiDAR for 3D depth.
- Led end-to-end integration of heater control subsystems (PID-based thermal regulation) within a safety-constrained electromechanical architecture, validating performance against IEC compliance standards.

Skills

Sensing & Instrumentation: Ultrasound, piezo sensing, EM tracking, IMU, ToF, LiDAR, sensor fusion, EKF, PyVISA/SCPI
Signal Processing & ML: Python (NumPy, SciPy), LightGBM, PyTorch, MATLAB/Simulink
Control & Embedded: PID, cascade/adaptive/robust control, MPC; C++, Embedded C, STM32
Hardware & Systems: PCB design (Altium, KiCad), prototyping, DFMEA, V&V, IEC 62368 / 60204